25900389 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCFSQ203 - FUNDAMENTALS OF FOOD SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Select the type of method used in mojonnier fat analysis [K] / [CO1]

   a). Gravimetric b). Colorimetric

   c). Titrimetric d). Enzymatic

2. Identify the conversion factor used in Kjeldahl method for crude protein from total nitrogen [U] / [CO1]

   a). 2.65 b). 4.5

   c). 5.55 d). 6.25

3. Select the primary reason for measuring moisture content in food products? [K] / [CO1]

   a). To determine the nutritional value b). To assess the physical properties of the food

   c). To ensure product quality, stability, and shelf life d). To measure the amount of fat in the food

4. Select the most common reagent used in Vitamin C estimation by titration [U] / [CO1]

a). Iodine solution b). Potassium permaganate

c). Nessler's reagent d). DCPIP

5. Recall the instrument used to measure refractive index: [K] / [CO1]

a). Spectrophotometer b). Polarimeter

c). Colorimeter d). Refractometer

6. Outline the type of reproduction in viruses [K] / [CO2]

a). Binary fission b). Asexual reproduction

c). Replication inside host cells d). Vegetative propagation

7. Recall the factor which does not directly affect microbial growth? [K] / [CO2]

a). Temperature b). pH

c). Osmotic pressure d). Type of microscope

8. State which of the following is absent in prokaryotic cells [K] / [CO2]

a). Nucleus b). Ribosome

c). Plasma membrane d). Cytoplasm

9. Select the immersion medium used with a 100x oil immersion objective lens: [U] / [CO2]

a). Water b). Oil

c). Alcohol d). Glycerol

10. State which is an example of food adulteration hazard [K] / [CO3]

a). Adding chalk powder to flour b). Presence of E.Coli

c). Glass fragment in salad d). Lead in water

[1x10 = 10]

Part B

Short Answer

Answer any **4** questions

Each question carries **5** marks

11. Define Wet Ash Analysis with its Procedure. [K] / [CO1]

12. Outline the Principle and Procedure of Gerber Method. [K] / [CO1]

13. Define Eukaryotes with iths characteristic features. [U] / [CO2]

14. Differentiate between Gram Positive and Gram Negative bacteria. [U] / [CO2]

15. Describe the role of maintenance in ensuring food safety and hygiene [U] / [CO3]

16. Explain on Personal Hygiene in food industry. [U] / [CO3]

[5x4 = 20]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

17. Explain in detail about Soxhlet method in fat analysis [U] / [CO1]

18. Define culture media. Describe in detail the different types of culture media with examples. [K] / [CO2]

19. Describe the streak plate,spread plate and pour plate methods [U] / [CO2]

20. Explain in detail on preproduction and post production operations in food industry [K] / [CO3]

[10x2 = 20]